DermAI: Skin Disease Detector from Mobile Images Using DL

1. Title

DermAI: Skin Disease Detector from Mobile Images Using Deep Learning

2. Objective

Build an AI-powered dermatological image analysis system that uses **mobile phone images of skin lesions and skin conditions** to automatically identify, classify, and analyze potential skin diseases using deep learning techniques.

The platform will allow users to capture or upload an image of a skin lesion through a mobile or web application. The system will preprocess the image, detect the relevant skin region, extract visual features, classify the condition using trained deep learning models, and provide an interpretable prediction with a confidence score.

The platform is designed to assist users, healthcare professionals, and researchers by providing an automated **preliminary skin-image analysis and decision-support mechanism**.

The solution focuses on:

- Mobile image acquisition
- Skin lesion/region detection
- Image quality assessment
- Image preprocessing
- Skin disease classification
- Multi-class disease detection
- Transfer learning and advanced CNN architectures
- Vision Transformer-based classification

- Confidence estimation
- Explainable AI visualization
- Prediction history
- Medical image analytics
- Automated analysis reports

The solution should be positioned as a **screening/research and clinical decision-support system**, not as an autonomous medical diagnostic system.

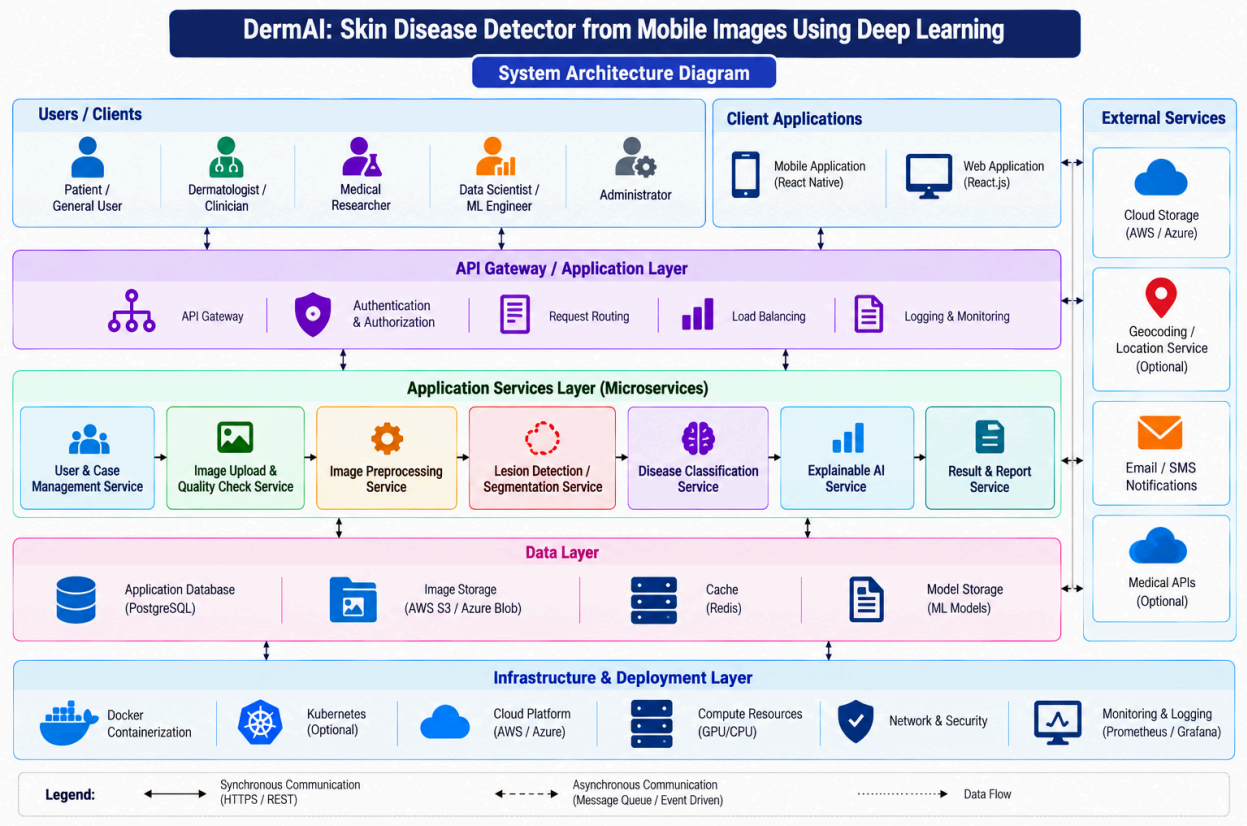
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3. Expected Outcomes

- Developed an AI-powered skin disease analysis platform.
- Implemented secure authentication and role-based access.
- Built mobile/web image upload and capture functionality.
- Implemented skin image quality assessment.
- Built image preprocessing and normalization pipelines.
- Implemented skin lesion detection/segmentation.
- Developed deep learning-based skin disease classification.
- Implemented multiple model architectures for comparison.
- Implemented severity/risk categorization where supported by the selected dataset.
- Added confidence scores to predictions.
- Implemented explainable AI visualizations.
- Built dashboards for users, clinicians, and administrators.
- Implemented prediction history and case management.
- Implemented automated analysis reports.
- Added model evaluation and comparison.
- Containerized the application using Docker.

- Deployed the application using cloud infrastructure such as AWS or Azure.

4. Architecture Diagram



5. Modules to be Implemented

1. User Authentication & Role-Based Access

The platform should provide secure authentication and authorization for different categories of users.

Features

- User registration
- User login
- JWT authentication

- OAuth2 authentication
- Role-based access control
- User profile management
- Secure password management
- Session management
- Audit logging

Roles

- **Patient / General User**
- **Dermatologist / Clinician**
- **Medical Researcher**
- **Data Scientist / ML Engineer**
- **Administrator**

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2. Patient & Case Management

This module manages skin-image analysis cases.

Patient Information

- Patient/Case ID
- Age
- Gender
- Relevant medical history
- Known allergies where applicable
- Clinical notes

Case Information

- Case ID
- Image ID

- Image capture date
- Body location
- Image source
- Analysis status
- Model version
- Prediction result

Case Management

- Create case
- Upload image
- Capture image
- View previous analyses
- Compare analysis results
- Track prediction history
- Add clinical notes

> For student implementation, real patient-identifiable information should not be used. Public/de-identified datasets should be preferred.

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3. Mobile Image Capture & Upload Module

This module handles images captured using mobile devices.

Image Acquisition

- Mobile camera integration
- Image upload
- Gallery selection
- Image preview
- Image cropping

- Image rotation
- Image orientation correction

Image Validation

- Supported image format validation
- Resolution validation
- File size validation
- Blurred-image detection
- Dark/overexposed image detection
- Image completeness validation

Image Quality Assessment

The system should determine whether the image is suitable for AI analysis.

Possible quality indicators:

- Blur level
- Brightness
- Contrast
- Resolution
- Focus quality
- Lesion visibility
- Background interference

Workflow

Mobile Camera → Image Capture → Quality Check → User Confirmation → Upload → AI Analysis

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4. Skin Image Preprocessing Engine

This module prepares the captured image for deep learning inference.

Image Processing

- Image resizing
- Pixel normalization
- Noise reduction
- Contrast enhancement
- Color normalization
- Image sharpening
- Background reduction
- Image standardization

Preprocessing Pipeline

Raw Mobile Image → Validation → Quality Assessment → Cropping → Resize → Normalization → Model Input

Additional Processing

- Data augmentation during training
- Color-space transformation
- Hair/artifact reduction where appropriate
- Region-of-interest extraction
- Image standardization

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5. Skin Lesion Detection & Segmentation Engine

This module identifies the relevant skin lesion or affected region before classification.

Segmentation Features

- Skin region detection
- Lesion localization
- Lesion boundary detection

- Lesion mask generation
- Background separation
- Multiple lesion detection

Candidate Models

- U-Net
- U-Net++
- Attention U-Net
- DeepLabV3+
- Mask R-CNN
- YOLO-based detection models
- Segmentation Transformer models

Segmentation Pipeline

Mobile Image → Preprocessing → Lesion Detection → Segmentation Model → Lesion Mask → ROI Extraction

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6. Skin Disease Classification Engine

This is the primary AI module responsible for classifying the skin condition.

Classification Features

- Multi-class skin disease classification
- Normal/abnormal classification
- Lesion category identification
- Probability estimation
- Top-N predictions
- Confidence scoring

Example Disease Categories

Depending on the selected dataset, categories may include:

- Melanoma
- Melanocytic nevus
- Basal cell carcinoma
- Actinic keratosis
- Benign keratosis
- Dermatofibroma
- Vascular lesion
- Other dataset-defined skin conditions

The actual disease classes must be determined by the **selected dataset and its ground-truth labels**.

Classification Pipeline

Skin Image → Preprocessing → Lesion ROI → Feature Extraction → Deep Learning Model → Disease Prediction → Confidence Score

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7. Advanced Deep Learning Intelligence Engine

This module provides the advanced AI capabilities of DermAI.

CNN-Based Models

- ResNet
- DenseNet
- EfficientNet
- MobileNet
- ConvNeXt

Transformer-Based Models

- Vision Transformer
- Swin Transformer

- DeiT

Advanced Techniques

- Transfer learning
- Fine-tuning
- Attention mechanisms
- Ensemble learning
- Feature fusion
- Multi-scale feature extraction
- Data augmentation
- Class imbalance handling
- Focal loss
- Weighted loss functions

Model Comparison

Multiple models should be trained and evaluated to identify the best-performing architecture for the selected dataset.

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8. Skin Disease Severity & Risk Assessment Module

Where the selected dataset provides appropriate labels or clinically validated categories, the system can provide an additional risk assessment layer.

Risk Categories

- Low Risk
- Moderate Risk
- High Risk
- Requires Clinical Review

Assessment Factors

- Predicted disease category
- Model confidence
- Lesion characteristics
- Image quality
- Historical analysis
- Dataset-supported clinical labels

The system should not independently invent clinical thresholds. Risk categories must be based on the **dataset labels, validated clinical criteria, or project-defined research protocol**.

Output

- Predicted condition
- Confidence score
- Risk category where supported
- Recommendation for professional evaluation where appropriate

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9. Explainable AI Module

Since the system analyzes medical images, model predictions should be accompanied by visual explanations.

Explainability Features

- Grad-CAM
- Grad-CAM++
- Attention maps
- Saliency maps
- Lesion segmentation overlay
- Prediction probability
- Top-N prediction comparison

Visualization

The user should be able to view:

Original Image + Lesion Region + AI Attention Map + Prediction

Example

```
Original Skin Image
        ↓
Lesion Region
        ↓
AI Attention Map
        ↓
Predicted Condition
        ↓
Confidence Score
```

The explanation should help users understand **which image regions influenced the model prediction**, while not being presented as proof of clinical correctness.

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10. AI Model Training & Evaluation Module

This module manages the complete model development lifecycle.

Training

- Dataset loading
- Dataset validation
- Train/validation/test splitting
- Data augmentation
- Class balancing
- Model training
- Hyperparameter tuning
- Checkpoint management
- Early stopping
- Learning-rate scheduling

Model Comparison

Models can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Sensitivity
- Specificity
- AUC-ROC
- Confusion matrix
- Balanced accuracy

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11. Image Quality Evaluation Module

Mobile images can vary significantly in quality, so image quality assessment should be treated as an independent component.

Quality Checks

- Blur detection
- Low-light detection
- Overexposure detection
- Underexposure detection
- Resolution check
- Occlusion detection
- Excessive background detection
- Lesion visibility

Output

The system can classify an uploaded image as:

- **Good Quality**
- **Acceptable Quality**
- **Poor Quality**

Poor-quality images can be rejected or sent back to the user for recapture.

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12. Classification Evaluation Module

This module evaluates how accurately DermAI classifies skin conditions.

Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Sensitivity
- Specificity
- AUC-ROC
- Balanced accuracy
- Matthews correlation coefficient

Detection Analysis

The system should compare:

Ground Truth ↔ Model Prediction

and identify:

- True Positive
- True Negative
- False Positive
- False Negative

Confusion Matrix

The system should provide class-wise confusion matrices to identify categories where the model performs poorly.

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13. Visualization & Analysis Dashboard

User Dashboard

- Recent analyses
- Previous predictions
- Prediction history
- Image results
- Confidence scores
- Reports

Clinician Dashboard

- Total cases
- Cases requiring review
- Disease distribution
- Prediction confidence
- Model performance
- Recent cases

Administrator Dashboard

- Registered users
- Total analyses
- Model usage
- System activity
- Prediction statistics

- Platform health

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14. Medical Image Analysis Screen

The analysis interface should display the complete AI prediction.

Display Components

- Original image
- Preprocessed image
- Lesion region
- Segmentation mask
- AI attention map
- Predicted condition
- Confidence score
- Top-N predictions
- Risk category where supported
- Model version
- Analysis timestamp

Comparison View

Users can compare:

Original Image ↔ Processed Image ↔ Lesion Mask ↔ Explainability Map

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15. Report Generation System

The platform should automatically generate an analysis report.

Report Contents

- Case ID
- Image information
- Analysis date

- Image quality result
- Detected lesion
- Predicted condition
- Confidence score
- Top-N predictions
- Risk category where supported
- Explainability visualization
- Model version
- Analysis summary

Export

- PDF
- JSON
- CSV

The report should clearly state that the output is **AI-assisted analysis and is not a standalone medical diagnosis**.

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16. Prediction History & Follow-Up Module

The system should maintain previous analysis records.

Features

- Analysis history
- Previous predictions
- Image comparison
- Date-based filtering
- Case tracking
- Follow-up analysis
- Model version tracking

Comparison

Users with appropriate permissions can compare images from different analysis sessions.

```
Previous Image
       ↓
Previous Prediction
       ↓
New Image
       ↓
New Prediction
       ↓
Analysis Comparison
```

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17. Notification & Alert System

The system can notify users about analysis results and follow-up requirements.

Notifications

- Analysis completed
- Poor image quality
- Low-confidence prediction
- High-risk prediction where supported
- Clinical review recommended
- Report generated
- Follow-up reminder

Alert Levels

- Informational
- Low
- Medium
- High

- Critical

Alert terminology should be carefully controlled because an AI prediction alone should not be represented as a confirmed diagnosis.

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18. Model Management & Versioning

This module manages deployed AI models.

Features

- Model registration
- Model versioning
- Model comparison
- Model deployment
- Model rollback
- Training history
- Dataset version tracking
- Performance tracking

Example

```
Model v1.0 → ResNet50
Model v1.1 → EfficientNet
Model v2.0 → DenseNet
Model v2.1 → Vision Transformer
Model v3.0 → Ensemble Model
```

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19. Final Integration, Testing & Deployment

Tasks

- Frontend-backend integration
- Mobile/web integration
- AI inference integration

- API validation
- End-to-end testing
- Image quality testing
- Model prediction testing
- Security testing
- Performance optimization
- Docker containerization
- Cloud deployment
- Monitoring and logging
- Documentation
- User guide preparation

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6. Week-wise Module Implementation and High-Level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Dataset & Core Setup

Tasks

- Define project objectives and workflow.
- Select target skin conditions.
- Select appropriate public/de-identified dataset.
- Understand dataset structure and annotations.
- Perform exploratory data analysis.
- Analyze class distribution.
- Design system architecture.
- Design database schema.

- Create UI wireframes.
- Set up frontend and backend.
- Implement authentication.
- Implement case management.
- Implement image upload.
- Build initial preprocessing pipeline.

Expected Outcome

- Project architecture completed.
- Dataset prepared.
- Dataset classes identified.
- Frontend and backend initialized.
- Authentication operational.
- Image upload operational.
- Basic preprocessing pipeline completed.

Recommended Dataset Categories

The exact dataset should be selected according to the target diseases and mobile-image use case.

Possible Dataset Sources

- Public dermoscopic image datasets
- Public skin lesion image datasets
- Public dermatology image datasets
- De-identified clinical skin-image datasets
- Institutionally approved datasets where available

The **specific dataset, disease categories, image modality, annotation type, and class distribution should be finalized before implementation**, because these

directly determine the classification methodology.

Milestone 2: Week 3 & 4 — Image Processing, Lesion Detection & Classification

Tasks

- Implement image quality assessment.
- Implement preprocessing pipeline.
- Implement lesion detection.
- Implement baseline classification model.
- Implement transfer learning.
- Train multiple CNN architectures.
- Handle class imbalance.
- Implement data augmentation.
- Generate prediction results.
- Compare model performance.
- Generate classification reports.

Expected Outcome

- Image quality module operational.
- Image preprocessing completed.
- Lesion detection operational.
- Skin disease classification operational.
- Multiple deep learning models evaluated.
- Best-performing baseline model selected.

Milestone 3: Week 5 & 6 — Advanced AI, Explainability & Risk Analysis

Tasks

- Implement advanced CNN architecture.
- Implement Transformer-based model.
- Implement ensemble model where appropriate.
- Optimize hyperparameters.
- Implement confidence estimation.
- Implement explainable AI.
- Generate Grad-CAM/attention visualizations.
- Implement risk classification where supported.
- Build medical image analysis dashboard.
- Implement prediction history.

Expected Outcome

- Advanced AI models operational.
- Model comparison completed.
- Explainable AI visualization completed.
- Confidence scoring implemented.
- Risk analysis implemented where supported.
- Analysis dashboard operational.

Milestone 4: Week 7 & 8 — Reporting, Testing, Analytics & Deployment

Tasks

- Build complete dashboard.
- Implement automated reports.
- Add PDF/CSV/JSON export.
- Implement prediction history.

- Implement notifications.
- Perform end-to-end testing.
- Perform API testing.
- Perform image-quality testing.
- Perform model inference testing.
- Perform security testing.
- Perform performance testing.
- Dockerize application.
- Deploy application.
- Configure monitoring and logging.
- Prepare final documentation.
- Prepare final presentation and demonstration.

Expected Outcome

- Fully integrated DermAI platform.
- Mobile/web image analysis workflow operational.
- Disease classification operational.
- Explainable AI operational.
- Reporting system operational.
- Dashboard completed.
- Application deployed.
- Complete end-to-end demonstration available.

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7. Evaluation Criteria

Milestone 1 — Week 2

- Project initialization completed.

- Dataset selected and analyzed.
- Disease classes identified.
- Database designed.
- Authentication implemented.
- Image upload operational.
- Preprocessing pipeline implemented.

Milestone 2 — Week 4

- Image quality assessment implemented.
- Lesion detection implemented.
- Classification model trained.
- Multiple models evaluated.
- Classification metrics calculated.
- Best-performing baseline model selected.

Milestone 3 — Week 6

- Advanced deep learning model implemented.
- Transformer/advanced architecture evaluated.
- Explainable AI implemented.
- Confidence scoring implemented.
- Risk analysis implemented where supported.
- Analysis dashboard completed.

Milestone 4 — Week 8

- Complete frontend/backend integration.
- Mobile/web image analysis demonstrated.
- End-to-end prediction workflow demonstrated.
- Reports generated.

- Testing completed.
- Docker deployment completed.
- Final documentation completed.

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8. Tools & Technology Stack

Programming Language

Python

Backend

- FastAPI
- REST APIs
- Pydantic
- SQLAlchemy

Frontend

- React.js
- JavaScript/TypeScript
- Tailwind CSS
- Chart.js / Plotly

Mobile Application

- React Native
- Expo
- Camera API

Database

Primary Database

- PostgreSQL

Optional

- MongoDB
- Redis

Image Storage

- AWS S3
- Azure Blob Storage
- MinIO for local development

AI & Deep Learning

- PyTorch
- TensorFlow
- MONAI where applicable
- Scikit-learn
- NumPy
- Pandas
- OpenCV
- Pillow

Classification Models

- ResNet
- DenseNet
- EfficientNet
- MobileNet
- ConvNeXt
- Vision Transformer
- Swin Transformer

Lesion Segmentation

- U-Net
- U-Net++
- Attention U-Net
- DeepLabV3+
- Mask R-CNN

Explainable AI

- Grad-CAM
- Grad-CAM++
- Captum

Medical Image Processing

- OpenCV
- Pillow
- NumPy
- scikit-image

MLOps

- MLflow
- Docker
- Docker Compose
- GitHub Actions

Cloud & Deployment

- AWS / Azure
- Docker
- Cloud storage

- Cloud database

Development Tools

- VS Code
- Git
- GitHub
- Postman
- Jupyter Notebook

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9. Performance Metrics

Image Quality Metrics

- Image quality classification accuracy
- Blur detection accuracy
- Image rejection rate
- Acceptable image percentage
- Poor-quality image detection rate

Disease Classification Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Sensitivity
- Specificity
- Balanced accuracy
- AUC-ROC
- Matthews correlation coefficient

Class-Level Metrics

- Per-class precision
- Per-class recall
- Per-class F1-score
- Confusion matrix
- One-vs-rest AUC

Lesion Segmentation Metrics

Where segmentation is implemented:

- Dice Similarity Coefficient
- IoU
- Pixel Accuracy
- Sensitivity
- Specificity
- Hausdorff Distance

Explainable AI Metrics

- Localization consistency
- Attention-map quality
- Lesion-region overlap
- Explanation stability

System Performance

- Image upload time
- Image preprocessing time
- Model inference time
- API response time
- Dashboard loading time

- Report generation time
- Concurrent user capacity
- GPU utilization

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10. Example Quantitative Goals

Image Quality Assessment

- Reliably identify images unsuitable for analysis.
- Reduce predictions generated from severely blurred or poorly exposed images.
- Maintain consistent preprocessing across supported mobile devices.

Skin Disease Classification

Target strong classification performance using:

- High macro F1-score
- High balanced accuracy
- Strong sensitivity for important disease classes
- Low false-negative rate for high-priority classes
- Consistent performance across test data

The exact numerical targets should be established **after selecting the dataset and establishing a baseline**, rather than arbitrarily defining clinical performance thresholds.

Lesion Segmentation

Where segmentation labels are available:

- High Dice coefficient
- High IoU
- Low boundary error
- Consistent lesion localization

Explainability

- Attention maps should focus predominantly on clinically relevant image regions.
- Segmentation and attention visualizations should be available alongside predictions.
- Prediction explanations should remain consistent across repeated inference where applicable.

Platform Performance

- Low-latency inference for supported mobile image resolutions.
- Responsive dashboard.
- Reliable image upload and processing.
- Reproducible model predictions.
- Stable operation under concurrent users.